

Artificial Intelligence

Assignment 7



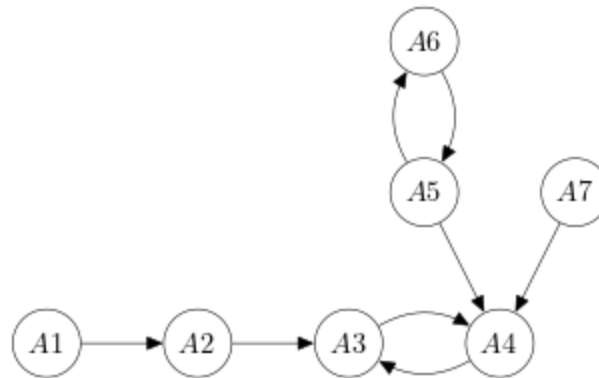
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1 Argumentation

(8 Points)

Assume the following abstract argumentation framework AF_1 :



For each of the following sets, decide whether they are (1) conflict-free and (2) admissible, and whether they are (3) preferred, and/or (4) grounded extensions of AF_1 . If they are not, give a short reason why not.

- a) $E = \{A_1, A_3, A_7\}$
- b) $E = \{A_1, A_4, A_5, A_7\}$
- c) $E = \{A_1, A_3, A_6\}$
- d) $E = \{A_1, A_3, A_5, A_7\}$

- a. $E = \{A_1, A_3, A_7\}$

It is conflict-free

It is admissible.

E is not a preferred extension as it is not a maximal set. To be a preferred extension it should support Maximality.

Grounded extension

- b. $E = \{A_1, A_4, A_5, A_7\}$

It is not a conflict-free

It is not admissible as it is not conflict-free

E is not a preferred extension as it is not not admissible as it is not conflict-free

As the set is not preferred, Grounded Extension is not applicable

- c. $E = \{A_1, A_3, A_6\}$

It is conflict-free

It is admissible

E is not a preferred extension as it is not a maximal set. To be a preferred extension it should support Maximality.

As the set is not preferred, Grounded Extension is not applicable

d. $E = \{A1, A3, A5, A7\}$

It is conflict-free

It is admissible.

E is preferred extension

Not a Grounded Extension

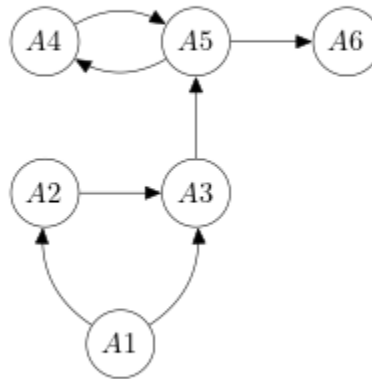
The maximal sets are : $\{A1, A3, A5, A7\}$ and $\{A1, A3, A6, A7\}$

Grounded Extension : $\{A1, A3, A7\}$

2 Argumentation

(5 Points)

Assume the following abstract argumentation framework AF_2 :



Find all complete and all stable extensions of AF_2 .



$$F_{AF}(\emptyset) = \emptyset$$

$$F_{AF}(\{A1\}) = \{A1\}$$

$$F_{AF}(\{A4\}) = \{A4\}$$

$$F_{AF}(\{A1, A4\}) = \{A1, A4\}$$

$$F_{AF}(\{A1, A5\}) = \{A1, A5\}$$

$$F_{AF}(\{A4, A6\}) = \{A4, A6\}$$

$$F_{AF}(\{A1, A4, A6\}) = \{A1, A4, A6\}$$

Complete Extension of AF2 : $\emptyset, \{A1\}, \{A4\}, \{A1, A4\}, \{A1, A5\}, \{A4, A6\}, \{A1, A4, A6\}$

Stable Extension :

$\{A1, A5\}, \{A1, A4, A6\}$ is the stable extension as it attacks every argument.